INCH POUND

MIL-C-3098/145A
23 December 1991
SUPERSEDING
MIL-C-3098/145
19 March 1982

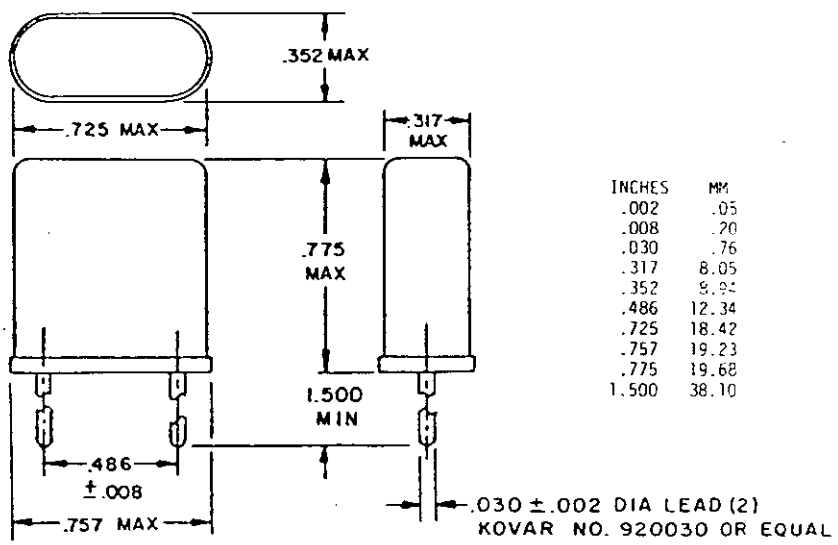
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR164/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; noncontrolled; series.



INCHES	MM
.002	.05
.008	.20
.030	.76
.317	8.05
.352	8.94
.486	12.34
.725	18.42
.757	19.23
.775	19.68
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The crystal unit shall not be trimmed after sealing.
4. Marking shall be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR164/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Holder: HC-33/U (MIL-H-10056/20).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 0.8 to 20 MHz, inclusive.

Frequency tolerance: ±50 ppm.

Equivalent resistance: See TABLE I.

TABLE I Equivalent resistance .

Frequency range, inclusive. MHz.			Maximum resistance Ohms
0.8	to	0.85	630
0.85+	to	0.9	600
0.9+	to	1	580
1+	to	1.12	540
1.12+	to	1.25	490
1.25+	to	1.37	450
1.37+	to	1.5	410
1.5+	to	1.62	380
1.62+	to	1.75	330
1.75+	to	1.87	300
1.87+	to	2.0	290
2.0+	to	2.12	270
2.12+	to	2.25	250
2.25+	to	2.6	200
2.6+	to	3	150
3+	to	3.4	110
3.4+	to	3.75	90
3.75+	to	4	75
4+	to	5	60
5+	to	7	35
7+	to	10	24
10+	to	15	22
15+	to	20	20

Mode of oscillation: Fundamental.

Operating temperature range: -55°C to +105°C, inclusive.

Drive level: 0.8 to 10.0 MHz = 10.0 ±2.0 mW.
10.0+ to 20.0 MHz = 5.0 ±1.0 mW.

Capacitance, shunt: 7.0 pF maximum.

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Shock:	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted	±10 ppm	±5 ppm
Equivalent resistance change permitted	±15 percent	±10 percent

Vibration: Method 201 Of MIL-STD-202.	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted	±10 ppm	±5 ppm
Equivalent resistance change permitted	±15 percent	±10 percent

Temperature cycling:	0.8 to 2.0 MHz	2.0+ to 20.0 MHz
Frequency change permitted	±10 ppm	±5 ppm
Equivalent resistance change permitted	±15 percent	±10 percent

Aging:
Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, ER, MI, SM
Navy - EC
Air Force - 85
DLA - ES

User activities:
Navy - AS, MC, SH
Air Force - 17, 19, 84

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5955-0659)

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